TQNT — Preregistration of dated predictions

(state as of 27 July 2026, before measurement)

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Purpose. This document freezes, *before measurement*, the dated quantitative predictions of TQNT-V6 (Topological Quantum Knot Theory, 6th edition), with their values, windows and *refutation conditions*. It is intended for a timestamped deposit (DOI) separate from the monograph, so that the future confrontation is accountable: any prediction listed here that is missed within its window counts *against* the theory, with no requalification possible. The values come from the TQNT-V6 monograph (version of 2026-07-27) and its scripts; none will be modified after deposit (any revision will go through a *new* dated version, the old one remaining accountable).

Declared antecedent. One prediction of the same programme is already *confirmed*: $\Omega_{cc} \sim 3\,727$ MeV (LHCb, 06/2026) within the predicted range $[3\,715, 3\,788]$ MeV. Honest chronology: the announcement preceded the final consolidation of the scale (a blind derivation in practice, not a formal preregistration) — hence precisely this document for what follows.

1 Hadronic spectroscopy (LHC, Run 3+)

Prediction	TQNT value	Refuted if
Ξ_{cc}^{*} (J=3/2 partner)	3 659 MeV	measured $\sim$ 3 690 (lattice/QM)
Ω_b^*	6 067 MeV (+22 vs Ω_b)	outside $\pm$ a few MeV
$\Omega_b(1P,\, j{=}0,\, J{=}\frac{1}{2})$ (the 5th of the quartet)	6 297–6 304 MeV	absence or outside window
B_0^*/B_1' (bottom j_K=1/2 doublet)	centroid 5 665 MeV (members $\sim$5 660/5 666, split 6.6 MeV)	5 591 or 5 517 (the $-\frac{1}{2}/-1$ readings); $\sim$ 5 700–5 750 (other models)
$\Sigma_b(1P)$ quintet	j_0 6 076; j_1 6 092/6 102; j_2 6 111/6 128 MeV	j_0/j_2 levels absent in $\Lambda_b\pi$
$\Sigma_c(1P)$ triplet	2 772/2 821/2 869 MeV ($\pm$ a few MeV)	outside windows
$\Xi_b'(1P,\, j{=}0)$	$\sim$6 243 MeV	— ($\Xi_b(6227)$ reservation declared)
Bottom 1D doublet	5 903/5 972 MeV (gap $\sim$69)	$B_J(5970) \neq$ 1D top ($J^P \neq 3^-$)
Ω_{ccc}	4 782 MeV (strict)	clear deviation
Ω_{bbb}	*outside the framework* (negative binding expected)	standard bound state found
H_{cc}^{++} (ground state 0^+)	4 497 $\pm$ 10 MeV, BOUND ($c\tau \sim 30\ \mu$m); channels $\Lambda_c^+ pK^-\pi^+$, $\Lambda_c^+\Lambda\pi^+$	no displaced-vertex signal in the listed channels
H_{cc} excited 1^+	4 572 $\pm$ 10 (pinned to the $\Lambda_c\Lambda_c$ threshold)	**any peak $>$ 4 573 MeV refutes**
H_{bb}^0 (0^+)	11 212 $\pm$ 10 MeV, bound ($c\tau \sim 220\ \mu$m)	—
Light H (dibaryon $uuddss$)	**ABSENT** (2 481, +250 above threshold)	any detection
Strange-baryon orbital bands	offset $s = 0.358(2)$ + full dilution	$s = 0.376$ (ΔWr/3)
j_K=3/2 splitting	$\kappa = 0.0650$ GeV^2 universal ($\times$0.813 strange)	non-universality
Z/H	exact degeneracy of the mirror doublets	any splitting (= CP-odd EW signature)

2 Families, neutrinos, weak currents

Prediction	Refuted if
Never a 4th family (derived Z_3 saturation)	any 4th generation
No sterile neutrino, at any mass	any confirmation (MicroBooNE, DUNE, KATRIN…)
$0\nu\beta\beta$: rate = 0 exactly (Dirac neutrinos)	any $0\nu\beta\beta$ signal
$\Sigma m_\nu \approx 66$ meV	very different cosmological value
Charged-current saturation: BR$_{\text{lep}}$/mode $\geq \sim$8 %	clear violation

3 Dark matter

Prediction	Refuted if
Absence in standard direct detection ($>$ 5 GeV, LZ/XENONnT)	standard WIMP detection
Absence also at 1.7–2.2 GeV (DS-20k, XLZD): $\sigma_{\text{eq}} \sim 10^{-48}$ cm^2	any detection in this window
6 narrow γ lines at 1.07/1.10/1.22/1.35/1.47/ 1.64 GeV ($\sim$1 dex below the 2015 limits; deep Fermi-LAT target)	smeared continuum at these energies (= tension); absence at $\times$10 sensitivities
No BH post-merger regenerated DM (P22 WKB)	residual post-merger signal

4 Cosmology and gravitation

Prediction	Refuted if
$\Omega_{\rm tot} = 1$ **exact** (flatness theorem)	any confirmed curvature
$w = -1$ **constant** ($dw/dz = 0$ by construction)	$w(z)$ confirmed (the DESI preference must *regress*)
H_0 and S_8 tensions: **must regress** (systematics/local void in GR)	new physics confirmed
Large-scale CMB anomalies: **no cross-replication** (E modes, lensing, 21 cm)	any cross-replication
Standard sirens at low $z \to$ global H_0	local amplification confirmed
Metrology of G: convergence **downwards** ($G_{\rm predicted} = 6.6710 \times 10^{-11}$)	consolidation at 6.674 (refutes at 5×10^{-4})
No GRB dispersion ($E_{\rm QG} > 9.3\,E_P$ protected)	dispersion detected
No drift of the constants (c, α: integer lock k=135)	confirmed drift
$r \leq 0.01$ (Starobinsky)	large r measured (LiteBIRD)

Accountability conditions

(i) The values above are frozen at the deposit date; (ii) a prediction missed within its window counts *against* the theory and will be recorded as such in the monograph (in the falsification register F1–F3, whose history shows the practice); (iii) *absence* predictions are refutable by any confirmed detection; (iv) any update will go through a new timestamped version, without withdrawal of the present one. Reference: TQNT-V6 monograph (associated Zenodo deposit), scripts of the software repository.